SPARKY'S SCIENCE EXPERIMENTS

Red Cabbage Colour Test: Acids and Bases

Physical Science (chemistry) · Ages 7+ · 30 minutes

Make purple red-cabbage juice that changes colour to show whether something is an acid or a base.

Safety first

Sparky always says: Never test bleach, drain cleaner or pool chemicals - they burn skin and eyes!

- An adult must use the knife and the hot water.
- Only test the kitchen things listed, and don't drink the mixtures.

You need

- A quarter of a red cabbage
- Hot water from the kettle (adult only)
- A jug, a sieve and 5 clear glasses
- To test: vinegar, lemon juice, bicarbonate of soda in water, plain water, soapy water

Steps

1. An adult chops the cabbage and pours hot water over it in the jug.
2. Wait 15 minutes until the water is dark purple, then pour it through the sieve.
3. Pour a finger-deep layer of the purple juice into each glass.
4. Add a teaspoon of a different thing to each glass. Stir, watch, and write down the colour.
5. Bonus: add vinegar to the bicarbonate of soda glass and watch it fizz!

If it doesn't work, try these in order

1. Colours weak? Use less water or more cabbage so the juice is darker.
2. No change? Add a little more of the thing you are testing and stir.

Why it works

Red cabbage has a natural purple colour (anthocyanin) that changes with what it is mixed with, so it is an indicator. Acids like vinegar and lemon juice turn it pink or red. Neutral things like plain water leave it purple. Bases like bicarbonate of soda and soap turn it blue or green. Vinegar (an acid) and bicarbonate of soda (a base) react and make carbon dioxide gas - the fizz.

Did you know? South African farmers test whether their soil is acidic, because mielies grow poorly in very acidic soil.

Ask Sparky: What colour will tomato juice make it? Is toothpaste an acid or a base?